

# Universal POWER SUPPLY

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**L**inear regulators like LM317, LM337, 78xx and 79xx, Zener diodes (1.3W) and reference diodes (e.g., LM336) are quite inexpensive and widely available. So you can use these liberally in power supplies.

Sometimes we need non-standard power supplies like  $\pm 11V$ ,  $+9V$  and  $-3V$ . This circuit shows how to get adjustable or fixed dual power supplies using LM317, LM337, 78xx and 79xx. The outputs can be regulated from practically any level to the maximum voltages of the installed voltage regulators. And there is no need to use symmetrical regulators; you can use LM317 and 7809, LM317 and LM337, 7809 and 7909, or 7809 and 7905, for instance.

As per application, the main difference between 78xx and 79xx and

LM317 and LM337 is their quiescent (bias) current. The quiescent current for LM317 or LM337 is around 0.1mA. For the fixed regulators, 78xx or 79xx, the quiescent current can reach 8mA. So the difference can be up to 80 times.

This circuit can be built with 78xx and 79xx, or with LM317 and LM337, so you should calculate values of the associated components accordingly before use. (You can even mix these, like LM317 for the positive output and 7905 for the negative output.)

## Circuit and working

Fig. 1 shows the circuit of a universal dual regulator with LM317 and LM337. The idea is to have adjustable output voltage from a predetermined reference voltage.

The positive output voltage  $V_1$  at CON3 (across points 1 and 2) is adjusted with potentiometer VR1. The negative output voltage  $V_4$  at CON3 (across points 2 and 3) is adjusted with potentiometer VR2.

Regulators IC1 and IC2 should be mounted on appropriate heat-sinks having thermal resistance, preferably, below ten degrees centigrade per watt.

You can use the circuit as a standard fixed power supply or as an adjustable power supply using 78xx and 79xx, or by using LM317 and LM337 with or without offset.

The components to determine before any application are: IC1, R4, ZD1, VR1 and R5 for the positive power supply and IC2, R3, ZD2, VR2 and R6 for the negative power supply. These components

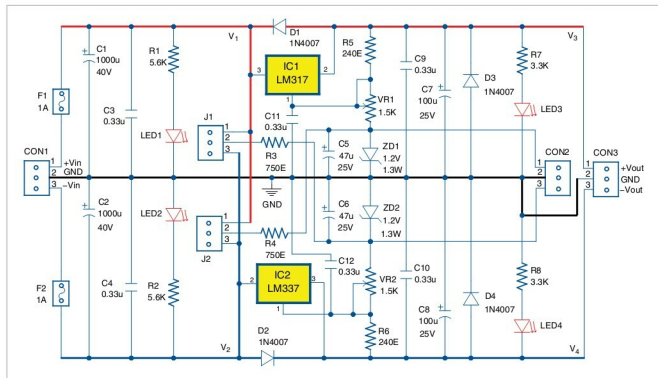


Fig. 1: Circuit diagram of the universal power supply